



Information processing mechanisms in microtubules at physiological temperature: Model predictions for experimental tests

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ABSTRACT

Both direct and indirect experimental evidence has shown signaling, communication and conductivity in microtubules (MTs). Theoretical models have predicted that MTs can be potentially used for both classical and quantum information processing although controversies arose in regard to physiological temperature effects on these capabilities. In this paper, MTs have been studied using well-established principles of classical statistical physics as applied to information processing, information storage and signal propagation. To investigate the existence of information processing in MTs we used cellular automata (CA) models with neighbor rules based on the electrostatic properties of the molecular structure of tubulin, and both synchronous and asynchronous updating methods. We obtained a phase diagram of possible dynamic behaviors in MTs that depend on the values of characteristic physical parameters that can be experimentally verified.

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1. Introduction

Microtubules (MTs) are cylindrical macromolecular structures composed of the protein tubulin. They are found in the cytoplasm of all eukaryotic cells and are the main component of the cell cytoskeleton. The main functions of MTs are (a) providing mechanical stability of the cell, (b) transporting cellular material via motor proteins, (c) moving the cell when forming the interior of cilia or flagella, and (d) separating chromosomes during mitosis (Dustin, 1978; Roberts and Hyams, 1979). More recently, MTs have been discovered to play a role in the communication between the cell exterior and the cell nucleus although the exact mechanisms of this process are not yet fully understood (Glanz, 1997; Maniotis et al., 1997). It has also been hypothesized that MTs are uniquely equipped to transmit signals within the cell (Gundersen and Cook, 1999) and models have shown the possibility of signal transfer via dipole interactions in axonal microtubules (Tuszynski et al., 1995; Brown and Tuszynski, 1997).

It has been speculated that MTs may potentially serve as information processing and transmission units within cells, and may function at both a classical and quantum level depending on architectural and environmental conditions. Theoretically the notion of quantum information processing has been strongly debated on all

sides (Tegmark, 2000; Hagan et al., 2002; Rosa and Faber, 2004), but much remains to be experimentally determined. Computer models of MTs as Ising spin systems (Tuszynski et al., 1998), quantum Hopfield neural networks (Behrman et al., 2006), and cellular automata (CA) (Hameroff et al., 1989) have been used to investigate these possibilities, and while they suggest the possibility of both classical and quantum information processing in MTs shielded from their noisy intracellular environment, they lack firm experimental parameters.

The purpose of this paper is to illustrate the importance of experimental parameters to this topic of interest. Section 2 will describe a two-state CA model based on previous CA (Hameroff et al., 1989) and ferroelectric models (Tuszynski et al., 1995, 1998; Brown and Tuszynski, 1997). Section 3 will outline the simulation conditions, physical parameters varied, and methods to evaluate the information processing capabilities of MTs at relevant biological temperature, and Section 4 presents the results of the simulations, placing them in the context of experimental investigation. Section 5 provides an overall summary of the paper and presents the authors' final conclusions.

2. CA Model of a Single MT

The tubulin molecule is a heterodimer composed of α -tubulin and β -tubulin. The α - β complex that assembles into MTs has dimensions of 8 nm $\times$ 5 nm $\times$ 4 nm (see Fig. 1). Dimers bind together end-to-end (i.e. β to α) in string-like structures known as protofilaments. MTs in human cells typically have 13 protofilaments (MT-13)

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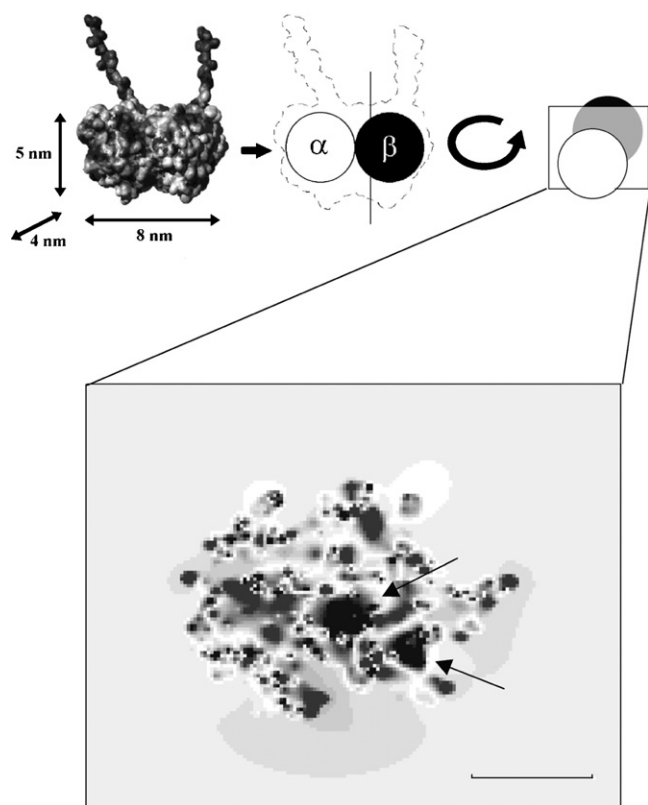


Fig. 1. Cross-section of an electrostatic map of the tubulin dimer showing the protein interior. Black arrows indicate regions of positive potential composing the double-well structure. Scale bar length 2 nm.

side by side that wrap to form a cylinder. The spacing between like monomers is given in Fig. 2. The slight shift between monomers in adjacent protofilaments results in a spiraling three-start left-handed helix of monomers. Two types of lattice structures exist. The A-lattice structure (MT-13A) consists of helices with alternating monomer types (i.e. α , β , α , β , α ... etc.) and possesses rotational symmetry and continuous wrapping, while the B-lattice structures (MT-13B) consist of helices with identical monomer types that change after one complete turn (i.e. 13 α , 13 β , 13 α ,... etc.) resulting in a MT with a physical discontinuity known as a seam.

Nogales et al. (1999) have shown that in general MTs possess parallel protofilaments limiting nearest neighbor configurations to an A-lattice with six neighbors (MT-13A-6) and a B-lattice with eight neighbors (MT-13B-8). Previous work by Campbell (2002) has

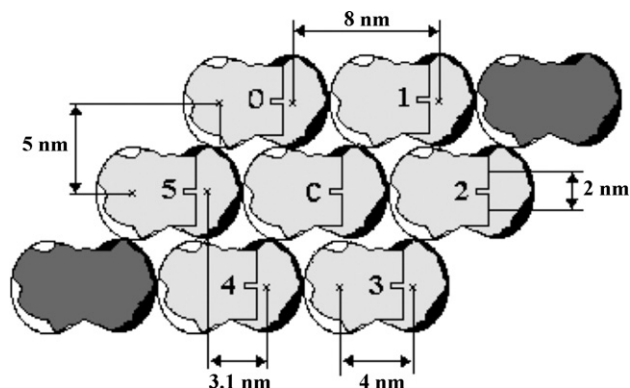


Fig. 2. Seven-dimer tilted hexagonal neighborhood of an individual tubulin dimer in a MT-13A-6 lattice structure showing dimensions of a MT-13A lattice with a double-well structure located in tubulin.

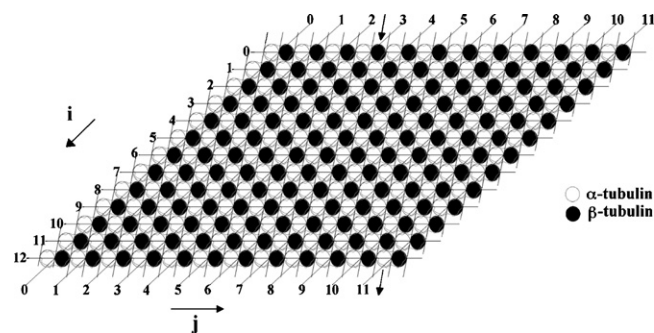


Fig. 3. Section of the two-dimensional array of tubulin dimers used to represent a MT-13A-6 lattice. Black arrows pointing along monomer helices indicate where monomer helices join between 0th and 12th protofilaments.

shown that the MT-13B-8 and MT-13A-6 configuration yield similar results in regards to its capability to process information. Thus, this investigation focuses on the MT-13A-6 configuration, with the assumption that similar results would be obtained for the MT-13B-8 lattice.

A single MT was modeled as a two-dimensional array measuring 13 rows and 100 columns (see Fig. 3) creating a MT-13A-6 of 0.8 μm in length. Dimer neighborhoods were constructed according to the tilted hexagonal shape of the MT-13A-6 structure (see Fig. 2). To create continuous boundary conditions the β ends of the dimer in the hundredth column were considered joined to the α end of the dimer in the first column, while neighbors 0 and 1 of dimers in the first row were located in the thirteenth row and neighbors 3 and 4 of dimers in the thirteenth row were in the first row in accord with the three-start helical winding pattern (see arrows in Fig. 3).

Solving the Poisson–Boltzmann equation for the 3.5 Å resolution crystallographic structure of the tubulin molecule (Lowe et al., 2001) via the Adaptive Poisson–Boltzmann Solver (APBS) software (Baker et al., 2001) produced an electrostatic potential map of the protein. By taking cross-sections of the resulting electrostatic map two regions of positive potential separated by 2 nm and surrounded by negative potential can be observed located near the separation between the α and β -tubulin monomers, approximately 4.5 nm from the tip of the α -monomer (see Fig. 1). This structure provides a local double-well potential for a mobile electron that may undergo an electron transfer (ET) process within the protein. This is consistent with the findings of Tuszyński et al. (2005) who found that the hydrophobic interior of the tubulin protein gives rise to a symmetric electrostatic potential, characteristic of a double-well structure indicating that nearby charged groups could execute tunneling motion between its energy minima.

Long-range electron tunneling appears to be a viable mechanism for conventional protein ET at physiological energies (Gray and Winkler, 2003). ET has been shown to appear as consecutive electron tunneling jumps between specific redox sites or protein-bound cofactors (Dreyer, 1984; Marcus and Sutin, 1985) with the typical distance between electron donors and acceptors being 10–15 Å (Canters and Dennison, 1995; Baum, 1993). However, the donor–acceptor interaction will be quite small when donors and acceptors are separated by distances greater than 10 Å (Gray and Winkler, 1996).

Hameroff et al. have examined possible conduction pathways for electron mobility and quantum tunneling among aromatic amino acids in tubulin by mapping intra-tubulin locations of tryptophan and histidine (Hameroff et al., 2002) as both are moderate electron donors (Pullman and Pullman, 1958). Overlays of the histidine, and tryptophan locations on the electrostatic map of tubulin revealed that histidine (β -28) (Lowe et al., 2001) is approximately 11 Å away from the right well of the double-well structure, and tryptophan (α -

407) (Lowe et al., 2001) is approximately 8 Å away from the left well. Thus, both histidine (β -28) and tryptophan (α -407) would be capable of supplying a mobile electron to the double-well structure for use in ET processes, although the histidine interaction will be quite small. Recent experimental evidence suggests that electron transfer within and among tryptophan residues may mediate protein function (Aubert et al., 2000; Wagenknecht et al., 2000), and Becker et al. (1975) have shown non-radiative photon exchange between tryptophan and aromatics in adjacent tubulins, and between MTs and membranes, suggesting that tryptophan has a large electron resonance in comparison to other aromatics. Therefore, tryptophan is considered as the most highly suited amino acid for transiting electrons and exchanging photons in MTs.

For the purpose of the model investigated here, the double-well structure of tubulin was approximated by an infinite square double-well potential described as,

$$\begin{aligned} V(x) &= +\infty & x \leq -L/2 \\ &= 0 & -L/2 < x < -a \\ &= V_0 & -a \leq x \leq a \\ &= 0 & a < x < L/2 \\ &= +\infty & x \geq L/2 \end{aligned} \quad (1)$$

where x is the position about an axis centered between the two wells within tubulin, L is the width of the double-well structure, $2a$ is the width of the potential barrier and V_0 is the height of the potential barrier. The dynamics of the system is based on the organization in time of mobile electrons located in the double-well structure in each of the tubulin dimers. Electrons were considered confined to individual tubulin dimers, but were allowed to move between the two potential wells localized within each dimer. The position of this mobile electron in the double well was used to designate the state of the tubulin cell. An electron in the left well designates the tubulin as being in an α -state since the left well is positioned closer to the α -tubulin monomer. An electron in the right well designates a β -state.

The Coulomb interaction was considered for electrons in nearest neighbor tubulins. While the Poisson–Boltzmann method discussed above yields an electrostatic map of an individual tubulin molecule in a solution of ions, there is little space between tubulin dimers polymerized into microtubules for ions and water and tubulins interacting via the lumen were not considered. As such, the energy, E , of a single electron in a central dimer C was determined by its Coulomb interaction with its neighbors as defined by,

$$E(C) = \sum_{m=0}^5 \frac{e^2}{4\pi\epsilon\epsilon_0 r(C, N_m)} \quad (2)$$

where e is the charge of the electron, ϵ is the dielectric constant of tubulin, ϵ_0 is the permittivity of free space, r_m is the separation between the central dimer electron and its m th neighbor electron N_m , and m is the index labeling the neighbors from 0 to 5 as shown in Fig. 2. To minimize the system's total energy, the energy difference between a central electron in an α -state and a β -state with the same configuration of neighbor electrons were compared. The energy difference, ΔE , is determined as,

$$\Delta E(C) = A \sum_{m=0}^5 \left(\frac{1}{r(C_f, N_m)} - \frac{1}{r(C_i, N_m)} \right) \quad (3)$$

where,

$$A = \frac{e^2}{4\pi\epsilon\epsilon_0} \quad (4)$$

The values for $E(C)$ and $\Delta E(C)$, are the two main parameters used to define the transition rules that govern the dynamics of the electrons in the MT system.

To noticeably alter the electrostatic map the movement of an electron in the double well would need to distort the map by more than the resolution of 3.5 Å. The majority of the charges defining the electrostatic map are bound charges, so the minimum work W required to noticeably distort the electrostatic map is given as,

$$\begin{aligned} W &= F\Delta x \\ &= \frac{1}{2} k\Delta x^2 \end{aligned} \quad (5)$$

where k is the tubulin dimer spring estimated as 4 N/m (Portet et al., 2005), and Δx is the distance by which the map is distorted, taken at a minimum of 3.5 Å. Thus, the minimum work required to distort the map is approximately 1.53 eV. Hameroff and Tuszynski (2003) have predicted the depth of the double-well potential as between 100 and 150 meV, thus under these conditions, the energy required for a single electron to move from one side of the well to the other is below 0.2 eV, well below the energy required to appreciably distort the electrostatic map. With the addition of surrounding dimers containing electrons, the interaction of electrons would raise an individual electron's energy. However, the interaction between dimers would also be expected to increase the spring constant due to the binding of the proteins and therefore is not expected to affect this result. Thus, the map shown above may be considered static for our purposes. As well, the maximum energy of an individual electron when interacting with surrounding electrons is approximately 0.6 eV as obtained from (2). While the energy gap to the conduction band in tubulin is unknown, a typical gap value between 2.57 and 3.12 eV has been reported for a number of proteins with 1.2 eV being the lowest value recorded for cytochrome C (Eley, 1989). As such it is reasonable to assume that the energies of the mobile valence electrons are insufficient to move the electrons to the conduction band in tubulin.

The mobile electron considered here is allowed to pass through the potential barrier via quantum tunneling with a finite probability. If the electron does not have enough energy to overcome the potential barrier, of height V_0 , then it can still make the transition with a tunneling probability defined by the parameters of the potential barrier between the two wells. The quantum solutions for the behavior of a wave packet in a one-dimensional infinite double square well potential separated by a finite barrier have been evaluated (Johnson and Williams, 1982; Razavy, 2003) and the time-dependent motion of a wave packet inside such a potential is also well studied (Johnson and Williams, 1982). Thus, the probability of finding a particle in one well or the other can be determined and used to evaluate the tunneling probability. These solutions to the time-dependent Schrödinger equations for a particle in a double well however, are solved for particles of constant energy. In this model the energy of an electron in a given dimer is determined by the relative positions of the electrons in the surrounding dimers. The positions of these electrons are updated at discrete time steps, and thus the energy of a given electron varies discretely with time. Therefore, there is difficulty in finding the tunneling probability due to the constant updating of the energy landscape at each time step. The most widely used approximation for solving tunneling problems is the semi-classical Wentzel–Kramers–Brillouin (WKB) method. The WKB method yields a tunneling probability P , dependent on the energy E of a particle, as,

$$P(E) = \frac{e^{-2\sigma}}{(1 + (1/4)e^{-2\sigma})^2} \quad (6)$$

with,

$$\sigma = \int_p^q \sqrt{\frac{2m}{\hbar^2} (V(x) - E)} dx \quad (7)$$

where p and q denote the classical turning points, and $V(x)$ is a potential that is a slowly varying function of x . When the condition $\sigma > 1$ is met, the tunneling probability is given by,

$$P(E) = e^{-2\sigma} \quad (8)$$

which agrees with the semi-classical approximation of Miller and Good (1953). It will be seen in Section 3 that this condition is met and that Eq. (8) can be used as a suitable approximation for the electron tunneling probability.

The tunneling transition does not depend on whether or not the energy of the electron will be minimized. That is,

- (i) for $E < V_0$ the transition probability is $P(E) = \exp(-2 \int_{-a}^a \sqrt{2m/\hbar^2 (V(x) - E)} dx)$, where $2a$ is the width of the potential barrier, and $\hbar$ is Planck's constant divided by 2π .

However, if the electron possesses enough energy to cross the potential barrier, and the transition between states lowers its energy, then it will switch states automatically. That is,

- (ii) for $E \geq V_0$ and $\Delta E \leq 0$ the transition probability is 1.

Rules (i) and (ii) instruct the CA model to update according to definite deterministic local rules. However, since the system is in contact with the thermal environment it is necessary to include local rules that incorporate thermal noise. As stated by Wolfram (1983) the simplest procedure is to impose that at each time step the position of the electron be switched with a probability P given by the Boltzmann factor for a finite temperature heat bath. This corresponds to cooler systems employing all the available energy to minimize their total energy, with hotter systems attempting the same but in the process allowing unfavorable transitions due to their interactions with the heat bath. This probability of the thermal energy is defined as $\exp(-\Delta E/kT)$, where k is Boltzmann's constant and T is the temperature of the system. Thus,

- (iii) for $E > V$ and $\Delta E > 0$ the transition probability is $P(E) = \exp(-\Delta E/kT)$.

These three rules (i)–(iii) govern the dynamics of the mobile electrons in the MT system.

3. Simulation: Conditions and Evaluation

There are several unknown parameters for the MT and its environment. The two main parameters varied during these simulations were the tubulin's potential well depth V_0 , and tubulin's dielectric constant ϵ . According to previous estimates (Hameroff and Tuszynski, 2003) V_0 was taken between 100 and 150 meV in 1 meV increments. With this range of well depths, and values of 2 nm and 2 Å for the well separation and the width of the potential barrier respectively (as determined from Fig. 1), it can be found from (7) that $\sigma > 1$ allowing the semi-classical approximation of (8).

For most proteins the dielectric constant is in the range of 2–4. However, recent experiments indicate that tubulin possesses a high-frequency dielectric constant of ~ 8.41 (Merishin et al., 2004), while others estimate the value between 1 and 100 (Minoura and Muto, 2006). It is doubtful that tubulin would have a dielectric constant greater than that of water, but it does indicate the difficulty in obtaining a precise value for tubulin's dielectric constant. To cover a plausible range of values the dielectric constant was taken between 2 and 10 in increments of 0.1.

The final factor that needed to be addressed was the method of updating. It should be noted that the method by which the states

change over time has significant impact on the behavior of CAs (Cornforth et al., 2005). In order to model a biological system realistically the method of updating should match the real-world system as much as possible. When a “global clock” is present, all CA cells should be updated synchronously. In the absence of a clock, cells should be updated in an asynchronous manner. Currently there is no direct experimental evidence suggesting the presence of a “global clock” affecting the MT system making it appropriate to use an asynchronous method of updating.

With this said, there are plausible “global clock” mechanisms that warrant consideration. The notion of coherent excitations supporting computation and information processing in MT has previously been suggested to occur via biological coherence (Hameroff, 1998). Biological coherence, as proposed by Fröhlich, is a high state of order that occurs in a biological system when the thermal energy supplied to the system reaches a certain threshold causing the coupled molecules to vibrate in unison (Fröhlich, 1968a,b). Fröhlich showed that when the energy supply to a system is sufficiently large in comparison with the energy loss, the dipoles tend to oscillate in a coherent manner. With increased energy supply from an external source, such as thermal fluctuations, deformations of the system caused by non-linear effects reduce the energy loss, moving the system into a metastable stationary state, such that the energy of the electric oscillations is larger than that in thermal equilibrium. Provided that the energy supply exceeds a critical value the excess energy of the system is channeled into a single longitudinal oscillation. Thus, the random energy supply is not converted wholly into thermal energy, but is used to maintain a coherent electric wave in the material. Fröhlich indicated that the channeling of randomly supplied energy into a single oscillation in biological systems is a special case of long-range quantum mechanical phase correlation, or coherence, such as in a Bose–Einstein condensate. The quantum properties exhibited by such systems have been used to explain the synchronization of biological rhythms such as sleep cycles, and heart rhythms. Non-thermal radiation at 8.085 MHz has been observed to be emitted from microtubules (Pokorný et al., 2001), and while this does not necessarily indicate the formation of a Fröhlich condensate it has been suggested (Pokorný et al., 2001; Pokorný, 2003, 2004, 1999). Thus, Fröhlich coherence may serve as a viable clocking mechanism for information processing in MT.

Another point to consider is the existence of strong electric fields across cellular membranes. These fields are responsible for the maintenance of gradients of charged molecules and ions, which allow many cell functions, such as control over cell volume and the electrical discharge of nerve or muscle cells. While these fields are accepted biological facts, the strength and effect of electric fields within cells is still under investigation. Tyner et al. (2007) have recently used a novel technique involving nano-sized voltmeters to map cellular wide electric fields, and have found that electric fields from mitochondrial membranes penetrate much deeper into the cytosol than previously estimated, a distance on the order of micrometers. Thus, while the exact mechanisms of a “global clock” within a cell are currently unknown, the existence of internal electric fields indicates that this possibility cannot be ruled out. Additionally, it has been proposed that dendrite and neuron function are regulated by the propagation of ionic waves along the cytoskeleton (Priel et al., 2005, 2006). The propagation of signals via ionic waves as they pass along MTs in the cytoskeletal network may also serve as a mechanism to synchronize electron behavior. In these cases it would be appropriate to use a synchronous method of updating. Since these issues are not conclusively resolved both synchronous and asynchronous methods were applied.

In all simulations the lattice of dimers was set to an initial configuration of α -states with β -state dimers randomly seeded

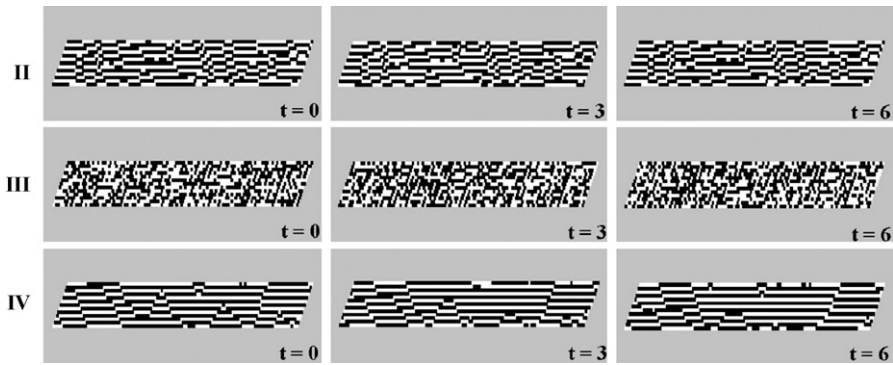


Fig. 4. A selection of output frames separated by three time steps showing the various automata behaviors exhibited during synchronous updating.

at a probability of 1%. The normal temperature of the environment was taken as 310 K. In the case of asynchronous updating, an independent random asynchronous method was used in which a time step consisted of randomly choosing an individual tubulin dimer on the MT lattice, with replacement, and changing the position of the electron according to the rules (i)–(iii) described above. Asynchronous simulations were run for 10^4 time steps to allow adequate time for patterns to develop. In the case of synchronous updating a time step consisted of systematically evaluating the preferred position of all the electrons in the lattice according to (i)–(iii) while holding all electrons in their original position, and then moving all the electrons simultaneously to their new positions. Synchronous simulations were run for 300 time steps.

Progression of the system from the initial random configuration was visually observed to determine the type of pattern developed. The patterns were classified using the following scheme developed in detail by Wolfram for the progression of one and two-dimensional CA (Wolfram, 1983, 1984; Packard and Wolfram, 1985).

Type I:	Progression leads to a homogenous state.
Type II:	Progression leads to a set of simple stable or periodic structures.
Type III:	Progression leads to an unpredictable pattern.
Type IV:	Progression leads to complex localized structures, sometimes long-lived.

Information in CAs is represented by dynamic, self-organizing patterns that are generated through neighboring cell interactions, indicating that Type IV behavior is characteristic of systems capable of processing information.

4. Results and Discussion

In the asynchronous simulations progression from the initial random configuration displayed constant change with no observable pattern formation over all ranges of V_0 and ϵ resulting in a Type III classification. Variations in the temperature around 310 K also did not affect this result. This suggests that in the absence of a clocking mechanism there is no capability of information processing within MTs at physiologically relevant temperatures.

Over the ranges of V_0 and ϵ synchronous updating lead to Types II–IV behaviors (see Fig. 4). Type II behavior consists of the entire system of dimers oscillating in unison between two opposite patterns, which was formed from the initial random configuration. Type III behavior displayed constant change and no observable pattern formation. Type IV behavior resulted in the formation of a rippling pattern moving over the microtubule on a background of oscillating bands.

In the synchronous simulations the overall behavior did not vary greatly as temperature was varied. However, as the temperature

was increased, the time taken to reach the corresponding asymptotic behavior did vary. Temperature increase delayed the onset of Types II and IV synchronous patterns, whereas it accelerated the onset of Type III behavior.

In all synchronous cases the typical behavior patterns were reached well within the 300 time-step period. If the energy of the electrons is taken on the order of ~ 100 meV, and the distance between wells as 2 nm, the time interval for a single transition would be on the order of femtoseconds. Thus, all behaviors for synchronous updating are reached on the timescale of picoseconds. However, this is assuming the movement of the electron is through a vacuum. Taking into account the medium may increase this timescale by several factors. Assuming transitions on the order of femtoseconds indicates that the movement of the Type IV bands around the MT occurs at speeds of $\sim 10^7$ m/s.

Fig. 5 gives the behavior of the synchronous system in terms of the tubulin parameters at a temperature of 310 K, which is relevant to the temperature of the typical microtubule environment. It should be noted that the boundaries are not firm but rather indicate transition regions between the three types of behavior. Type II behavior is characteristic of automata cells that are too strongly interdependent, while Type III behavior is characteristic of cells with too weak interdependence. Type IV behavior approaching from the Type II region is marked by the oscillating bands of Type II slowly varying to give rise to the Type IV discontinuities described. These discontinuities allow the formation of moving

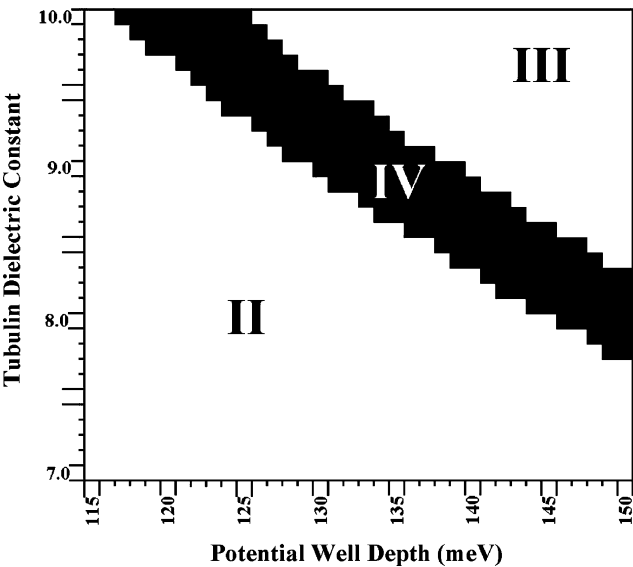


Fig. 5. CA behavior for synchronous updating based on tubulin dielectric constant versus tubulin potential well depth for a system temperature of 310 K.

objects and increase in size and movement in the further regions of Type IV behavior. As the region of Type III behavior is neared the moving objects become larger and interact giving rise to more complex behavior. Eventually this results in the loss of observable patterns giving rise to Type III behavior. Interestingly, below a tubulin dielectric constant of 7.8, independent of the potential well depth, only Type II behavior was observed giving an empirically testable hypothesis. This was also observed for barrier heights below 116 meV regardless of the dielectric constant value. The behavior of the system with a tubulin dielectric constant of 7.8 or higher and barrier height above 116 meV can be seen to depend on both the dielectric constant and the potential well depth.

The stable and oscillating patterns of MT automata observed via the synchronous simulations could possibly function as specific sites of ion or protein binding and transport thus performing some role in the orchestration of biomolecular activities. It is expected that the addition of microtubule associated proteins (MAPs) and other proteins known to normally interact with MTs will affect the overall patterns of behavior by altering the dynamics of electrons at specific protein attachment sites. As well, these simulations have shown that information processing at temperatures relevant to the MT environment is feasible so long as a clocking mechanism is present. The notion of information processing and signaling suggested to occur in MTs, at relevant temperatures, thus remains a possibility and is open to further investigation. Additionally, nanotech uses of biomolecules are of interest and the MT is a candidate for use in nanoscale devices.

The results of these simulations depend heavily on the electrostatic properties of the tubulin dimer and on the nature of the MT and its environment. However, there are no concrete experimental findings and ultimately experimental confirmation is needed to establish the origin and properties of an electron transfer process in MTs. Unfortunately, conductance measurements on biopolymers are fraught with difficulties due to their structural heterogeneity and instability, the liquid state of samples and the dependence of biological systems on environmental factors such as pH, temperature, and ionic concentrations. Despite these difficulties, a number of experiments have investigated the intrinsic conductivities of MTs (Fritzsch et al., 1998; Goddard and Whittier, 2006; Umnov et al., 2007).

In regards to the intrinsic conductivity values, which are relevant to the findings in this paper, Fritzsch et al. (1998) used a substrate embedded with gold microelectrodes to make electrical contacts to single dry MTs and found a resistivity value of approximately $40 \text{ M}\Omega/\mu\text{m}$. The dry state of the MTs however, brings the value into question when the cellular environment is considered. Using RF reflectance spectroscopy measurements Goddard and Whittier (2006) have taken various measurements of tubulin in a buffer solution in the presence and absence of MT associated proteins (MAPs). An average resistance of 883Ω was reported for MTs in a buffer solution in the absence of MAPs. The translation of this value into properties for individual MTs is not direct without assumptions about the MT geometry and connectivity. Assuming that all tubulin are polymerized with MT uniformly distributed in combinations of parallel and series networks, and assuming MT were $\sim 10 \mu\text{m}$ long Tuszyński et al. (2007) used the above findings to estimate a longitudinal MT resistivity of $0.8 \text{ M}\Omega/\mu\text{m}$. This predicts MT resistances two orders of magnitude smaller than dry MTs, which is to be expected. More recently Umnov et al. (2007) attempted direct measurement of MT conductivity by placing a solution of MTs in microchannels. They determined a 90 S/m upper bound on the conductivity of a MT, corresponding to a minimum $240 \text{ M}\Omega$ for a $10 \mu\text{m}$ MT ($24 \text{ M}\Omega/\mu\text{m}$), close to the Fritzsch et al. (1998) result. Although this research does not directly address the issues raised in this paper they do provide an experimental starting point from which they may be resolved.

5. Summary and Conclusion

The possibility of quantum processes in relation to information processing in MTs at physiologically relevant temperature has been investigated via a cellular automata computer model of a MT. The model uses a typical MT configuration of 13 protofilaments with its constituent tubulin proteins packed into a seven-member neighborhood in a tilted hexagon configuration known as an A-lattice. Based on previous results it is assumed that results for the A-lattice may also be extended to the B-lattice configuration. Solutions of the Poisson–Boltzmann equation for the crystallographic structure of tubulin revealed that the interior of the protein contains a region of two areas of positive charge separated by a barrier of negative charge. This interior arrangement was taken to constitute a double-well potential structure, within which a mobile electron may move, thus determining the electronic state of the tubulin dimer. The α -407 tryptophan residue is suggested as a likely donor for this mobile electron. Dynamics of the system are determined by the minimization of the overall energy associated with the electrostatic interactions between electrons in nearest neighbor tubulin dimers, as well as thermal effects based on Boltzmann factors. The model allows transitions for electrons with sufficient energy to overcome the potential barrier in which the new configuration lowers the system's energy, or if the configuration raises the system's energy with a finite probability, and allows the electron to tunnel through the potential barrier allowing transitions, at a finite probability, for which the system's energy is lowered even if the electron does not possess the necessary energy to overcome the potential barrier. To cover plausible ranges the system was investigated at 310 K with a potential barrier height between 100 and 150 meV, and a tubulin dielectric constant between 2 and 10. The system was updated in both a synchronous and asynchronous manner to consider both the presence and absence of a "global clock". Visual observation of the system revealed no capacity for information processing in the absence of a "global clock", while information processing in the presence of a clocking mechanism is possible for a narrow range of barrier heights and specific values for the tubulin dielectric constant. Thus, it is concluded that, based on a mobile electron in a double-well structure in tubulin, a clocking mechanism is required for physiologically relevant information processing in MT, and is only possible if the tubulin dielectric constant is above 7.8 with the potential barrier height exceeding 116 meV.

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References

- Aubert, C., Vos, M.H., Mathis, P., Eker, A.P.M., Bretel, K., 2000. Intraprotein radical transfer during photoactivation of DNA photolyase. *Nature* 405, 586–590.
- Baker, N.A., Sept, D., Joseph, S., Holst, M.J., McCammon, J.A., 2001. Electrostatics of nanosystems: application to microtubules and the ribosome. *Proc. Natl. Acad. Sci. U.S.A.* 98, 10037–10041.
- Baum, R.M., 1993. Views on biological, long-range electron transfer stir debate. *Chem. Eng. News* 22, 20–23.
- Becker, J.S., Oliver, J.M., Berlin, R.D., 1975. Fluorescence techniques for following interactions of microtubule subunits and membranes. *Nature* 254, 152–154.
- Behrman, E.C., Gaddam, K., Steck, J.E., Skinner, S.R., 2006. MTs as a quantum Hopfield network. In: Tuszyński, J.A. (Ed.), *The Emerging Physics of Consciousness*. Springer, Berlin, Heidelberg, New York, pp. 351–370.
- Brown, J.A., Tuszyński, J.A., 1997. Dipole interactions in axonal microtubules as a mechanism of signal propagation. *Phys. Rev. E* 56, 5834–5840.
- Campbell, R.D.J., 2002. Information processing in microtubules. Dissertation. Queensland University of Technology.
- Canter, G.W., Dennison, C., 1995. Biological electron transfer: structural and mechanistic studies. *Biochimie* 77, 506–515.

- Cornforth, D., Green, D.G., Newth, D., 2005. Ordered asynchronous processes in multi-agent systems. *Phys. D* 204, 70–82.
- Dreyer, J.L., 1984. Electron transfer in biological systems: an overview. *Experientia* 40, 653–675.
- Dustin, P., 1978. *Microtubules*. Springer-Verlag, New York.
- Eley, D.D., 1989. Studies of organic semiconductors for 40 years – I The mobile π -electron – 40 years on. *Mol. Cryst. Liq. Cryst.* 171, 1–21.
- Fritzsche, W., Böhm, K., Unger, E., Köhler, J.M., 1998. Making electrical contact to single molecules. *Nanotechnology* 9, 177–183.
- Fröhlich, H., 1968a. Bose condensation of strongly excited longitudinal electric modes. *Phys. Lett. A* 26, 402–403.
- Fröhlich, H., 1968b. Long-range coherence and energy storage in biological systems. *Int. J. Quant. Chem.* 2, 641–649.
- Glanz, J., 1997. Force-carrying web pervades living cell. *Science* 276, 678–679.
- Goddard, G., Whittier, J.E., 2006. Biomolecules as nanomaterials: interface characterization for sensor development. *Proc. SPIE* 6172, 617206.1–617206.12.
- Gray, H.B., Winkler, J.R., 1996. Electron transfer in proteins. *Annu. Rev. Biochem.* 65, 537–561.
- Gray, H.B., Winkler, J.R., 2003. Electron tunneling through proteins. *Q. Rev. Biophys.* 36, 341–372.
- Gundersen, G.G., Cook, T.A., 1999. Microtubules and signal transduction. *Curr. Opin. Cell Biol.* 11, 81–94.
- Hagan, S., Hameroff, S.R., Tuszynski, J.A., 2002. Quantum computation in brain microtubules: decoherence and biological feasibility. *Phys. Rev. E* 65, 061901.1–061901.11.
- Hameroff, S., 1998. Quantum computation in brain microtubules? The Penrose–Hameroff ‘Orch OR’ model of consciousness. *Phil. Trans. R. Soc. Lond. A* 356, 1869–1896.
- Hameroff, S.R., Tuszynski, J.A., 2003. Search for quantum and classical modes of information processing in microtubules: implications for the living state. In: Musmeci, F., Brizhik, L.S., Ho, M. (Eds.), *Energy and Information Transfer in Biological Systems*. World Scientific, Singapore, pp. 31–62.
- Hameroff, S.R., Rasmussen, S., Mansson, B., 1989. Molecular automata in microtubules: basic computational logic of the living state. In: Langton, C. (Ed.), *Artificial Life*, Santa Fe Institute Studies in the Science of Complexity, vol. VI. Addison-Wesley, New York, pp. 521–553.
- Hameroff, S., Nip, A., Porter, M., Tuszynski, J., 2002. Conduction pathways in microtubules, biological quantum computation, and consciousness. *Biosystems* 64, 149–168.
- Johnson, E.A., Williams, H.T., 1982. Quantum solutions for a symmetric double well. *Am. J. Phys.* 50, 239–243.
- Lowe, J., Li, H., Downing, K.H., Nogales, E., 2001. Refined structure of β -tubulin at 3.5 Å resolution. *J. Mol. Biol.* 313, 1045–1057.
- Maniotis, A.J., Chen, C.S., Ingber, D.E., 1997. Demonstration of mechanical connections between integrins, cytoskeletal filaments, and nucleoplasm that stabilize nuclear structure. *Proc. Natl. Acad. Sci. U.S.A.* 94, 849–854.
- Marcus, R.A., Sutin, N., 1985. Electron transfers in chemistry and biology. *Biochim. Biophys. Acta* 811, 265–322.
- Mershin, A., Kolomenski, A.A., Schuessler, H.A., Nanopoulos, D.V., 2004. Tubulin dipole moment, dielectric constant and quantum behavior: computer simulations, experimental results and suggestions. *Biosystems* 77, 73–85.
- Miller Jr., S.C., Good Jr., R.H., 1953. A WKB-type approximation to the Schrödinger equation. *Phys. Rev.* 91, 174–179.
- Minoura, I., Muto, E., 2006. Dielectric measurements of individual microtubules using the electroorientation method. *Biophys. J.* 90, 3739–3748.
- Nogales, E., Whittaker, M., Milligan, R.A., Downing, K.H., 1999. High-resolution model of the microtubule. *Cell* 96, 79–88.
- Packard, N.H., Wolfram, S., 1985. Two-dimensional cellular automata. *J. Stat. Phys.* 38, 901–946.
- Pokorný, J., 1999. Conditions for coherent vibrations in the cytoskeleton. *Bioelectrochem. Bioenerg.* 48, 267–271.
- Pokorný, J., 2003. Viscous effects on polar vibrations in microtubules. *Electromagn. Biol. Med.* 22, 15–29.
- Pokorný, J., 2004. Excitation of vibrations in microtubules in living cells. *Bioelectrochemistry* 63, 321–326.
- Pokorný, J., Hasek, J., Jelínek, F., Saroch, J., Palán, B., 2001. Electromagnetic activity of yeast cells in the M phase. *Electro-Magnetobiology* 20, 371–396.
- Portet, S., Tuszynski, J.A., Hogue, C.W.V., Dixon, J.M., 2005. Elastic vibrations in seamless microtubules. *Eur. Biophys. J.* 34, 912–920.
- Priel, A., Tuszynski, J.A., Woolf, N.J., 2005. Transitions in microtubule C-termini conformations as a possible dendritic signalling phenomenon. *Eur. Biophys. J.* 35, 40–52.
- Priel, A., Tuszynski, J.A., Cantiello, H.F., 2006. The dendritic cytoskeleton as a computational device: an hypothesis. In: Tuszynski, J.A. (Ed.), *The Emerging Physics of Consciousness*. Springer, Berlin, Heidelberg, New York, pp. 293–325.
- Pullman, B., Pullman, A., 1958. Electron-donor and -acceptor properties of biologically important purines, pyrimidines, pteridines, flavins, and aromatic amino acids. *Proc. Natl. Acad. Sci. U.S.A.* 44, 1197–1202.
- Razavy, M., 2003. *Quantum Theory of Tunneling*. World Scientific, River Edge.
- Roberts, K., Hyams, J.S., 1979. *Microtubules*. Academic Press, New York.
- Rosa, L.P., Faber, J., 2004. Quantum models of the mind: are they compatible with environment decoherence? *Phys. Rev. E* 70, 031902.1–031902.6.
- Tegmark, M., 2000. Importance of quantum decoherence in brain processes. *Phys. Rev. E* 61, 4194–4206.
- Tuszynski, J.A., Hameroff, S., Sataric, M.V., Trpisova, B., Nip, M.L.A., 1995. Ferroelectric behavior in microtubule dipole lattices: implications for information processing, signaling and assembly/disassembly. *J. Theor. Biol.* 174, 371–380.
- Tuszynski, J.A., Brown, J.A., Hawrylak, P., 1998. Dielectric polarization, electrical conduction, information processing and quantum computation in microtubules. Are they plausible? *Phil. Trans. R. Soc. A* 356, 1897–1926.
- Tuszynski, J.A., Brown, J.A., Crawford, E., Carpenter, E.J., Nip, M.L.A., Dixon, J.M., Sataric, M.V., 2005. Molecular dynamics simulations of tubulin structure and calculations of electrostatic properties of microtubules. *Math. Comput. Model.* 41, 1055–1070.
- Tuszynski, J.A., Priel, A., Brown, J.A., Cantiello, H.F., Dixon, J.M., 2007. Electronic and ionic conductivities of microtubules and actin filaments, their consequences for cell signaling and applications to bioelectronics. In: Lyshevski, S.E. (Ed.), *CRC Nano and Molecular Electronics Handbook*. Taylor and Francis, London, 18–1–18–46.
- Tyner, K.M., Kopelman, R., Philbert, M.A., 2007. Nanosized voltmeter enables cellular-wide electric field mapping. *Biophys. J.* 93, 1163–1174.
- Umnov, M., Palusinski, O.A., Deymier, P.A., Guzman, R., Hoying, J., Barnaby, H., Yang, Y., Raghavan, S., 2007. Experimental evaluation of electrical conductivity of microtubules. *J. Mater. Sci.* 42, 373–378.
- Wagenknecht, H.A., Stemp, E.D.A., Barton, J.K., 2000. Evidence of electron transfer from peptides to DNA: oxidation of DNA-bound tryptophan using the flash-quench technique. *J. Am. Chem. Soc.* 122, 1–7.
- Wolfram, S., 1983. Statistical mechanics of cellular automata. *Rev. Mod. Phys.* 55, 601–644.
- Wolfram, S., 1984. Universality and complexity in cellular automata. *Phys. D* 10, 1–35.